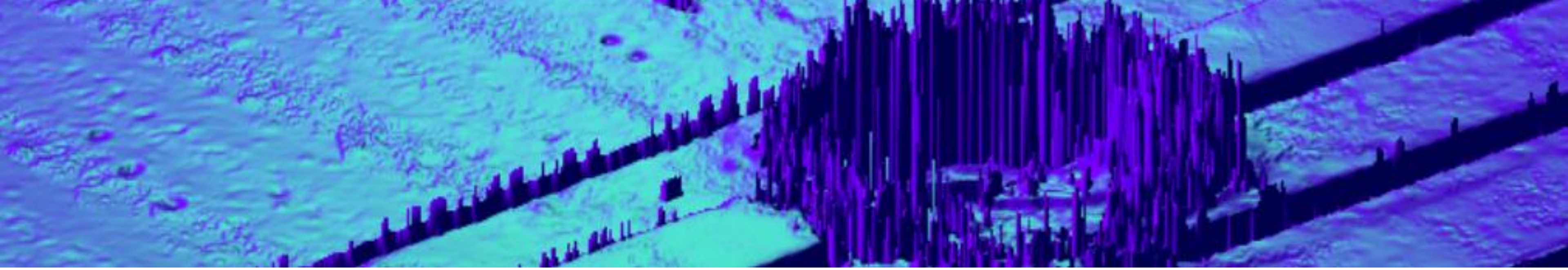

Engineered scattering for feedback suppression in 405 nm superluminescent diodes

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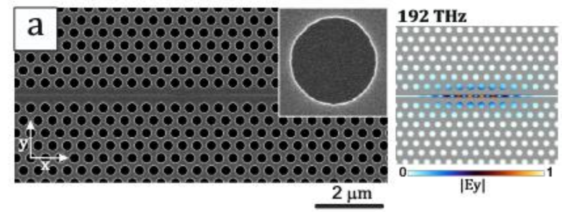
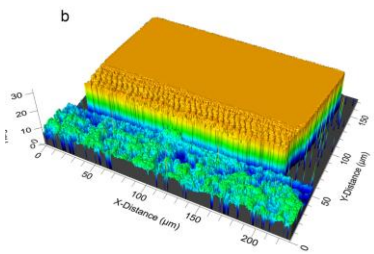
Our Team

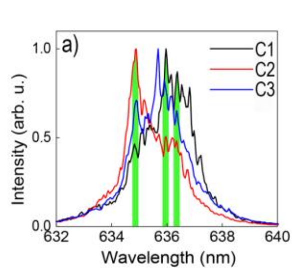
Photonics Crystal Group

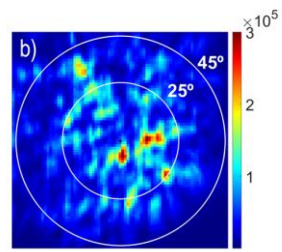
We study **disordered structures** in photonics devices, with application in **random lasing, opto-mechanics, radiative cooling** and **perovskite** emitters.

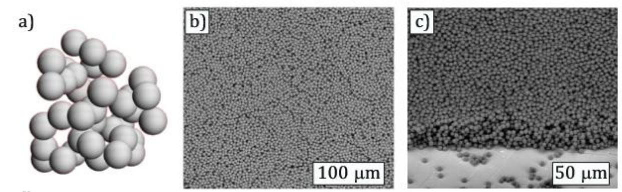


Phys. Rev. Lett. 130, 043802 (2023)

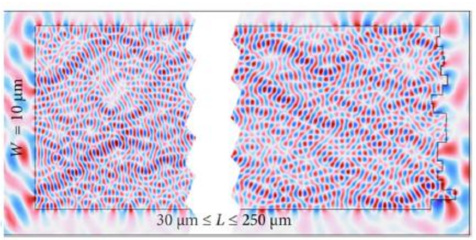


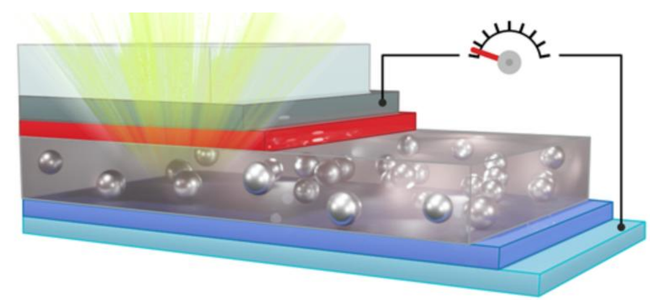






ACS Appl. Opt. Mater. 2, 896-897 (2024)





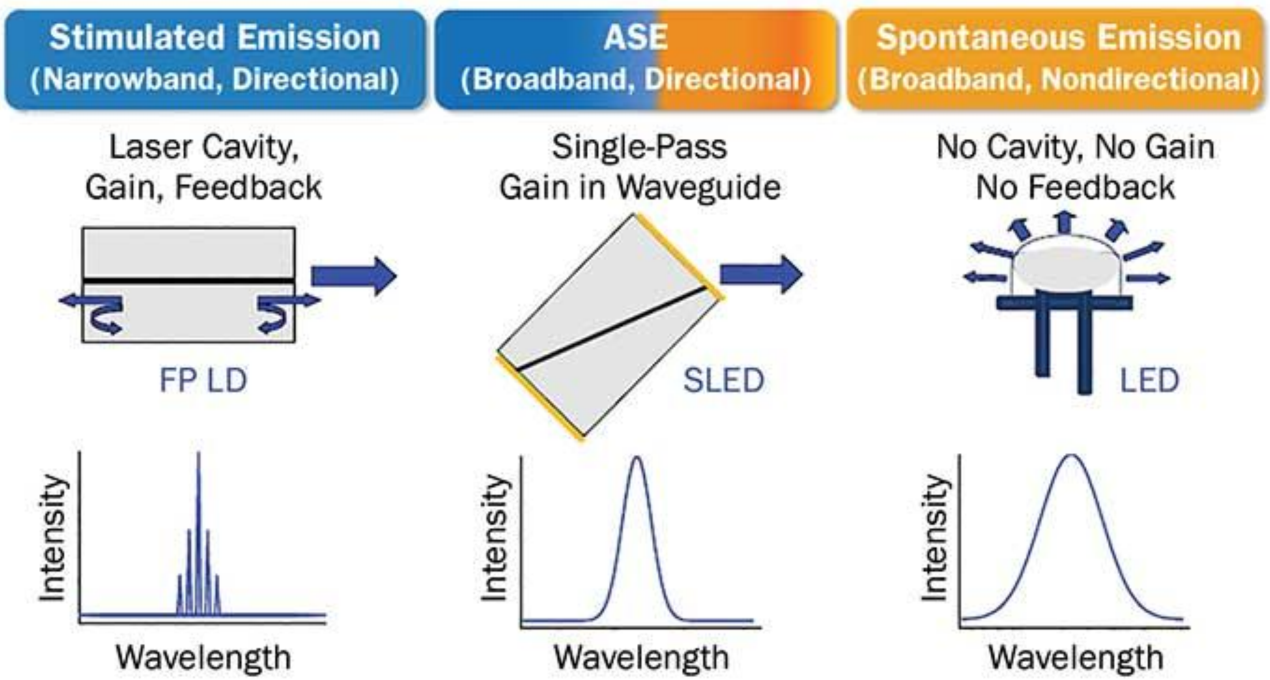
Nat. Photonics 16, 219–225 (2022)
Optica 11, 7, 919-925 (2024)
Nanophotonics 14, 27, 5375-5384 (2025)
J. Phys. Photonics 8 015059 (2026)

RSC Adv. 15, 39, 32497–32508 (2025)

Introduction

A Superluminiscent Diode (SLD or SLED) is a laser diode without feedback

Laser Diode | SLD | LED



Introduction

	Laser Diode	SLD	LED
Operating Principle	ASE + feedback	Amplified Spontaneous Emission (ASE)	Spontaneous Emission
Linewidth	0.1 - 1 nm	5 - 20 nm	20-50 nm
Spatial Coherence	High	Medium	Low
Div. Angle	5-30°	10-40°	120°

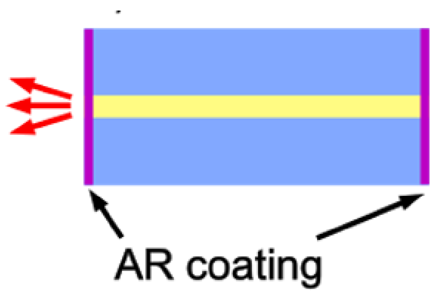
Applications where **high radiance** and **low spatial coherence** are needed:

- Optical Coherence Tomography
- Fiber Optics Gyroscopes
- Visible Light Communications
- Pico-projectors

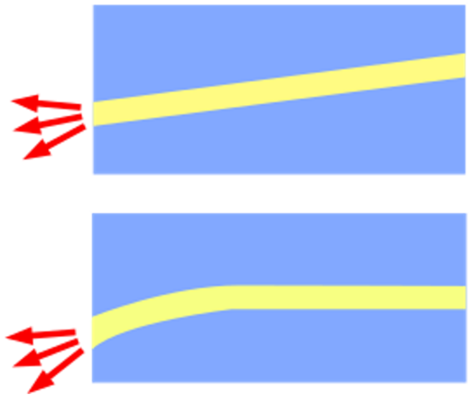
H. Lu et al. “Low-coherence semiconductor light sources: devices and applications” npj Nanophotonics 1, 9 (2024)

Introduction

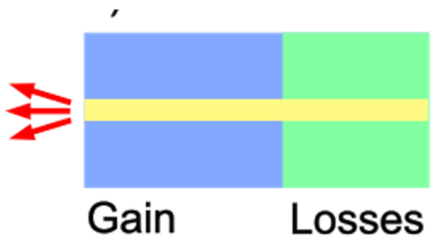
Anti Reflection coating



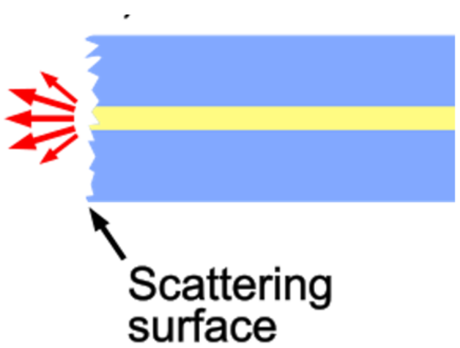
Tilting or bending



Additional absorber



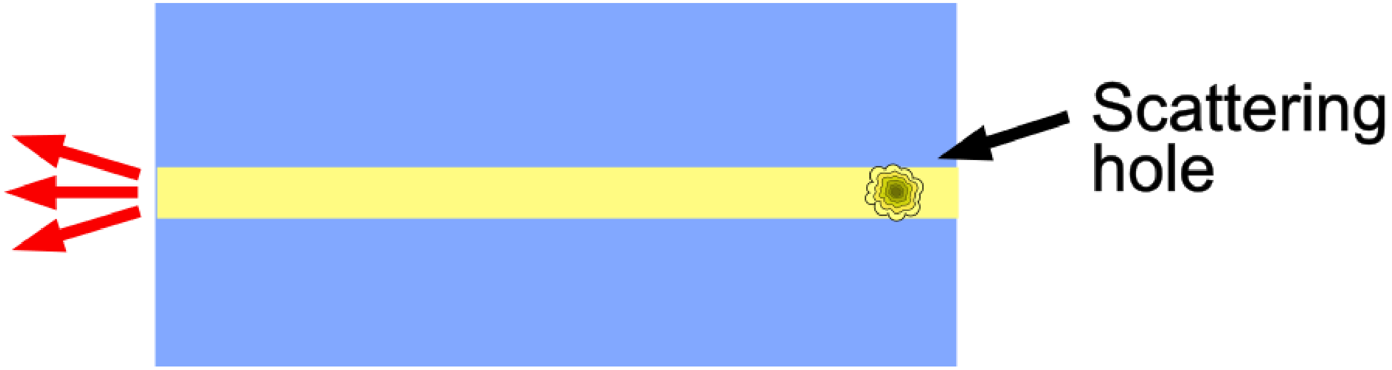
Roughen output mirror



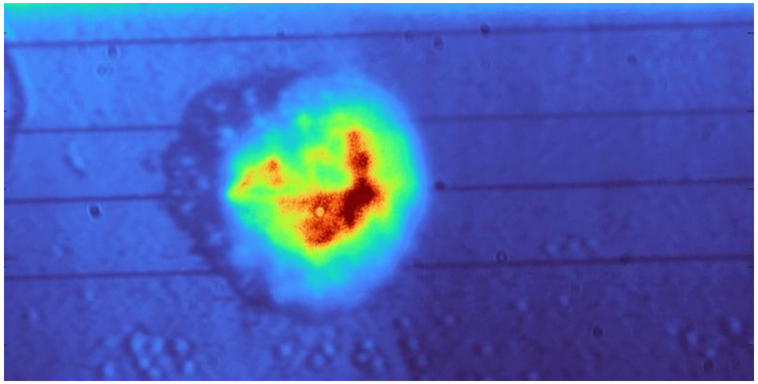
A. Kafar et al."Cavity suppression in nitride based superluminescent diodes" J. Appl. Phys. **111** 083106 (2012)
M. Hardy et al. "m-plane GaN-based blue superluminescent diodes fabricated using selective chemical wet etching" Appl. Phys. Express **2** 121004 (2009)

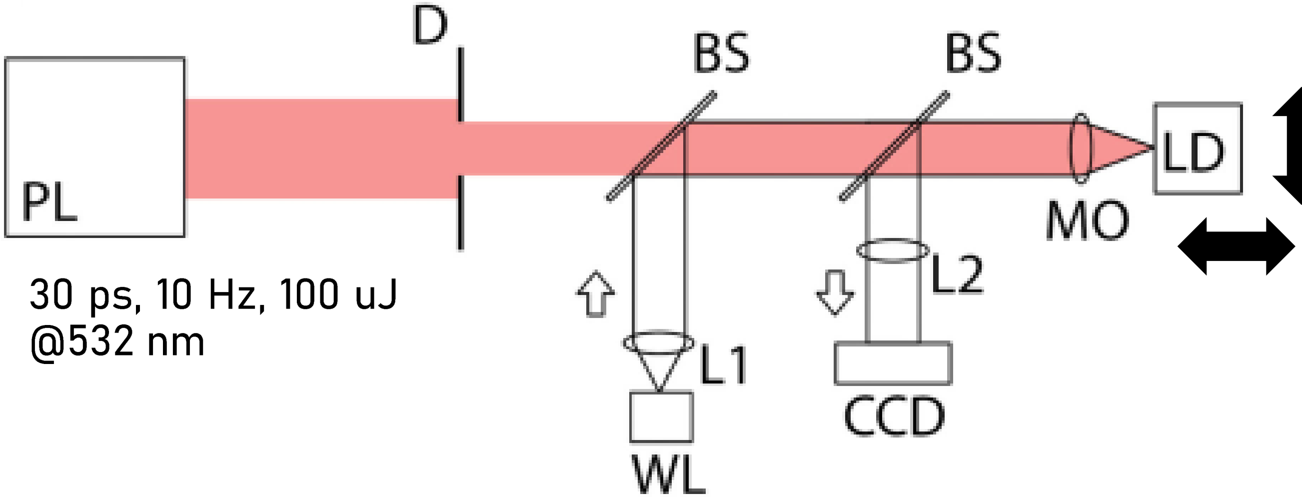
Proposed method

Post-process existing Laser Diodes, break the waveguide with scattering defects, maintaining uniform beam



Ablation set-up



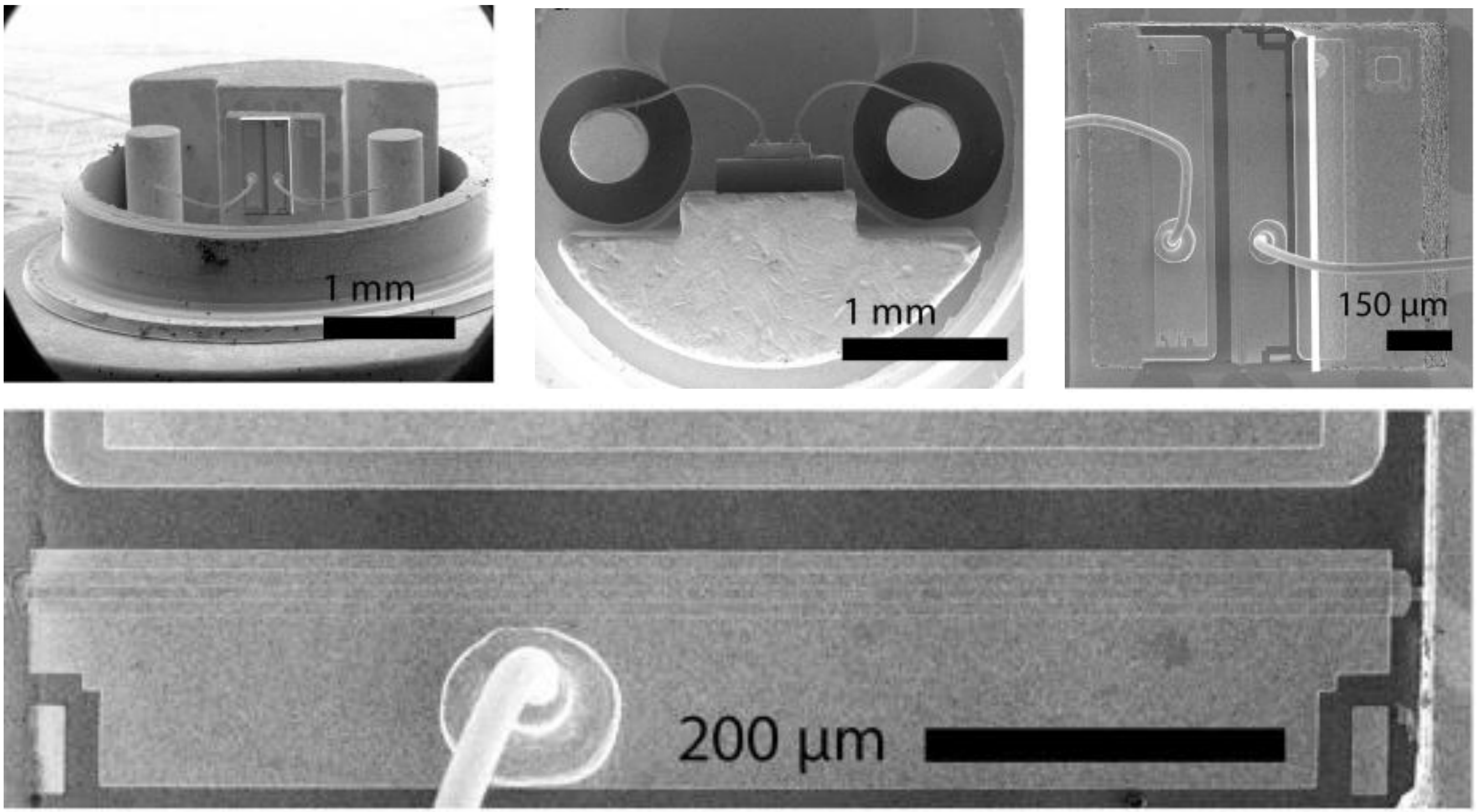


30 ps, 10 Hz, 100 uJ
@532 nm

Parameters:
- Spot diameter
- Energy pulse
- Number of delivered pulses
- Spot position

Original LD: structure

Fabry-Perot GaN Laser Diode, 405 nm, TO-can package, ridge waveguide (10 µm width, 750 µm length).



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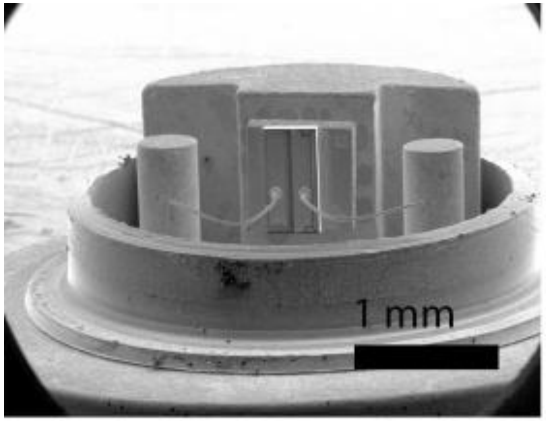
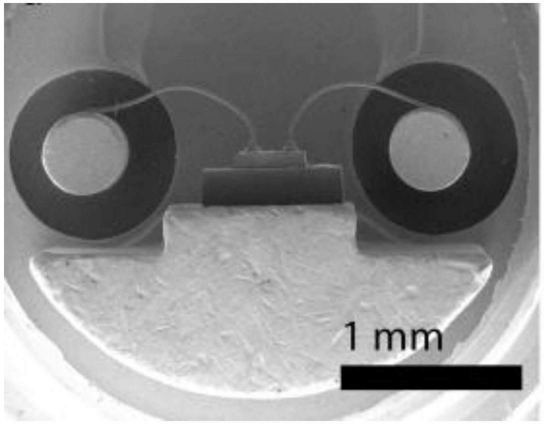
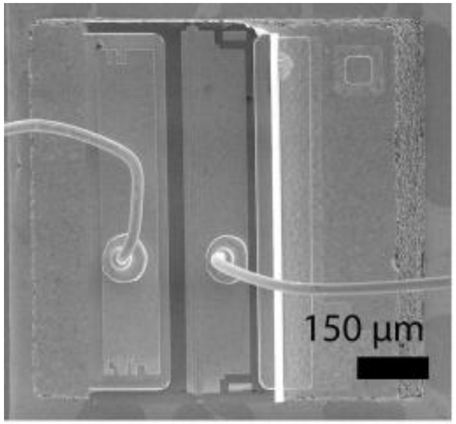
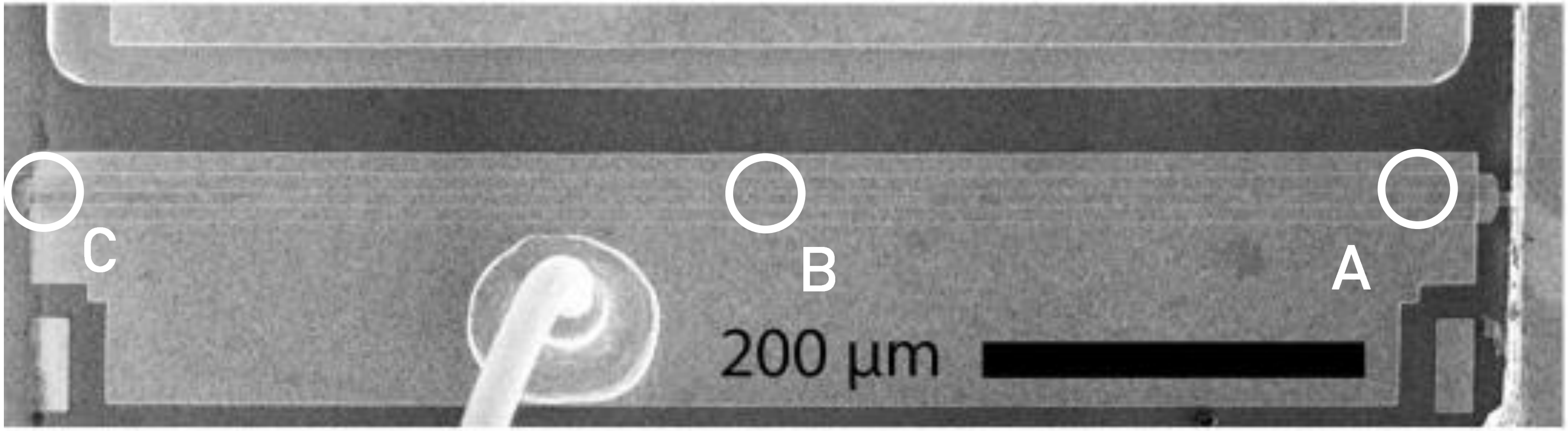


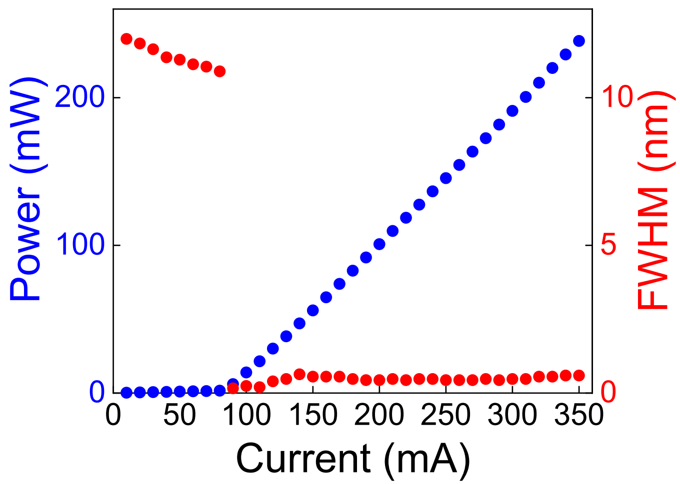


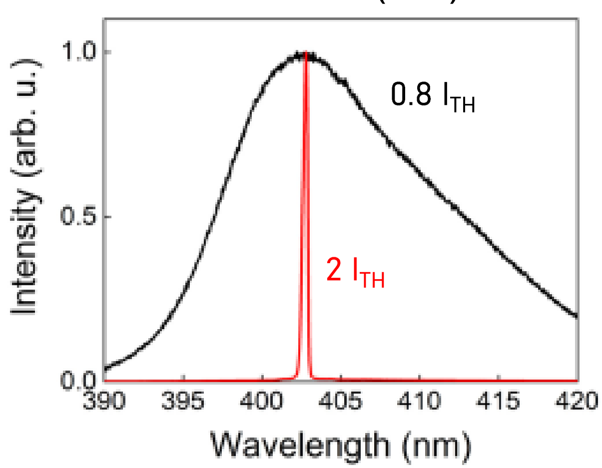


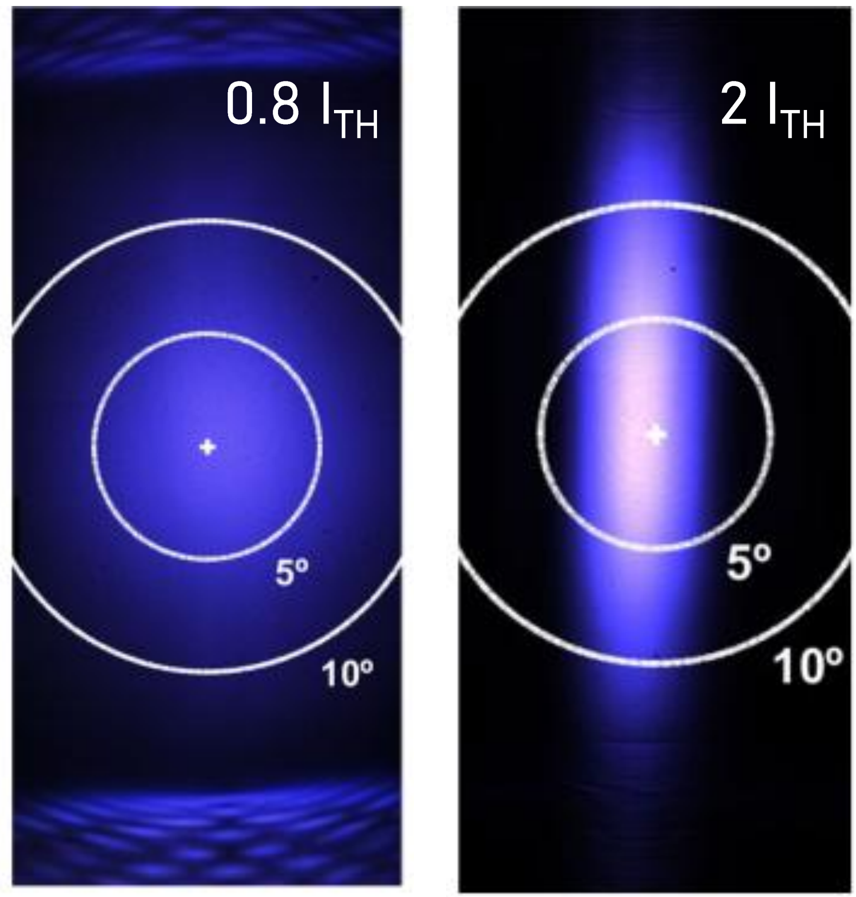


Original LD: emission

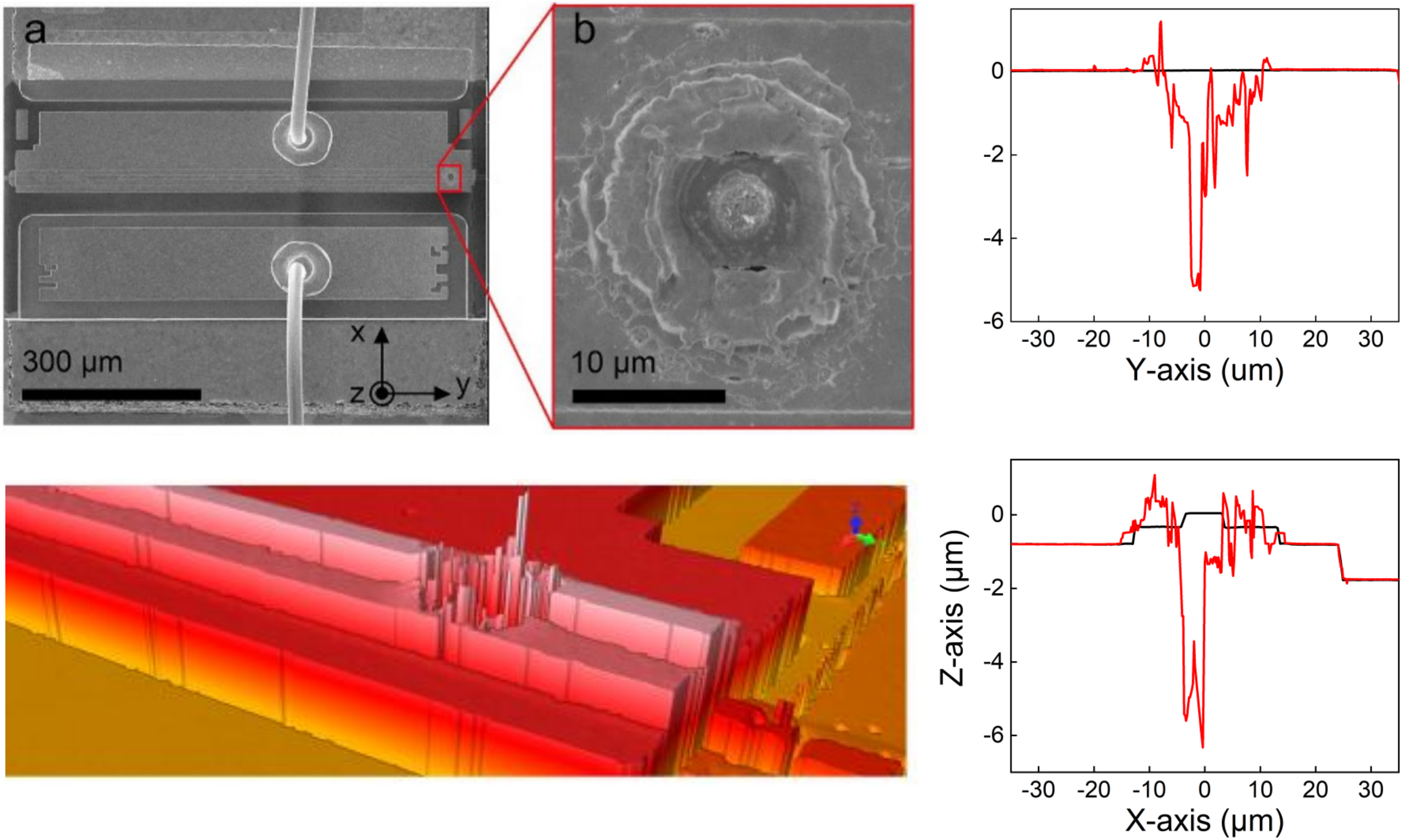
I_{TH} = 90 mA, P_{OUT} 250 mW at 350 mA, λ_C = 405 nm,
FWHM = 0.2 nm @ 2·I_{TH}, div. angles 6º (||) and 32º (⊥) @2 I_{TH}.







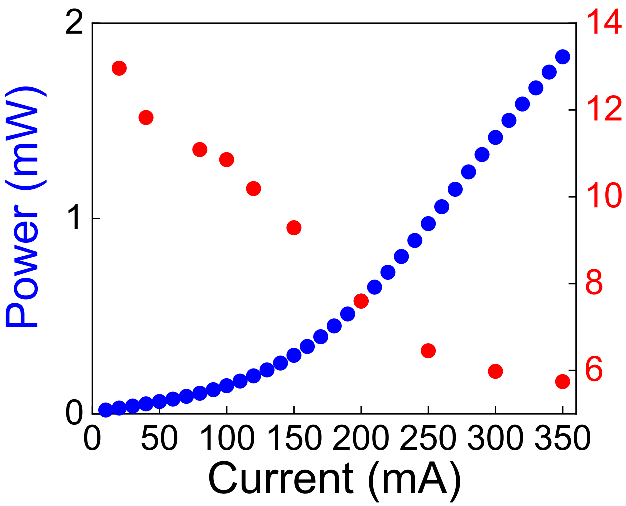
Results: device A structure

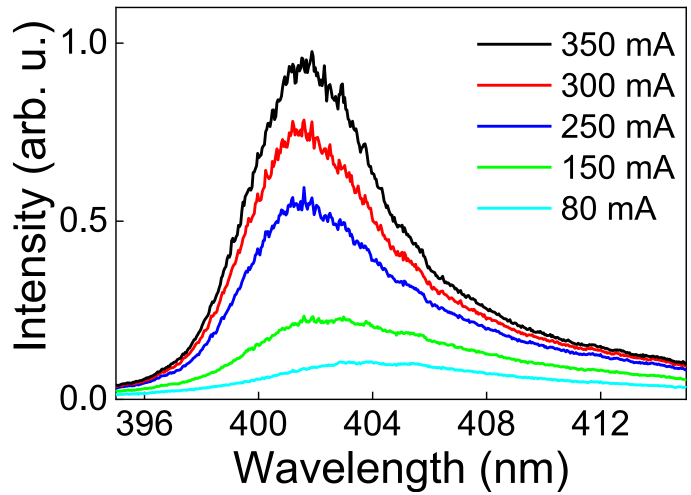


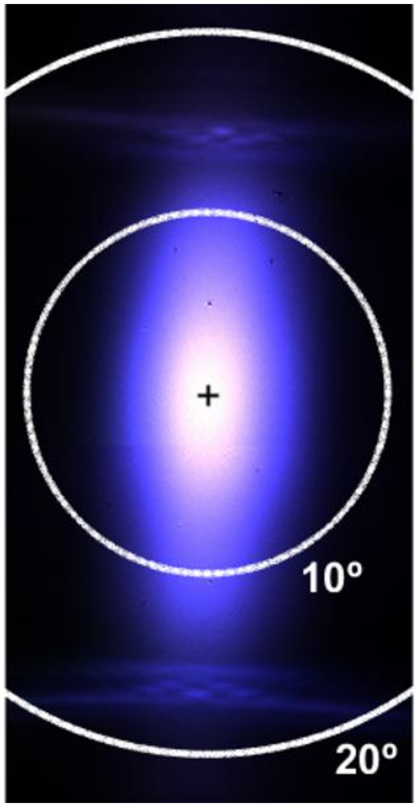
A. M. Pacheco, A. Consoli and C. López "Feedback suppression in 405 nm superluminescent diodes via engineered scattering" J. Phys. Photonics 8 015059 (2026)

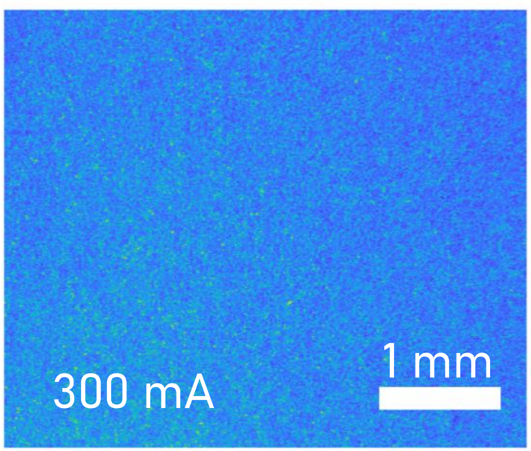
Results: device A emission

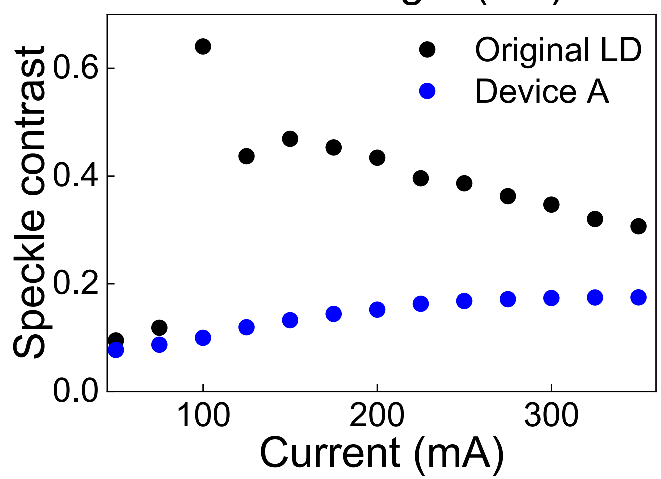
P_{OUT} = 1.8 mW, FWHM = 5.9 nm, div. angles 12º (∥) and 35º (⊥) at 350 mA. Lower speckle contrast.







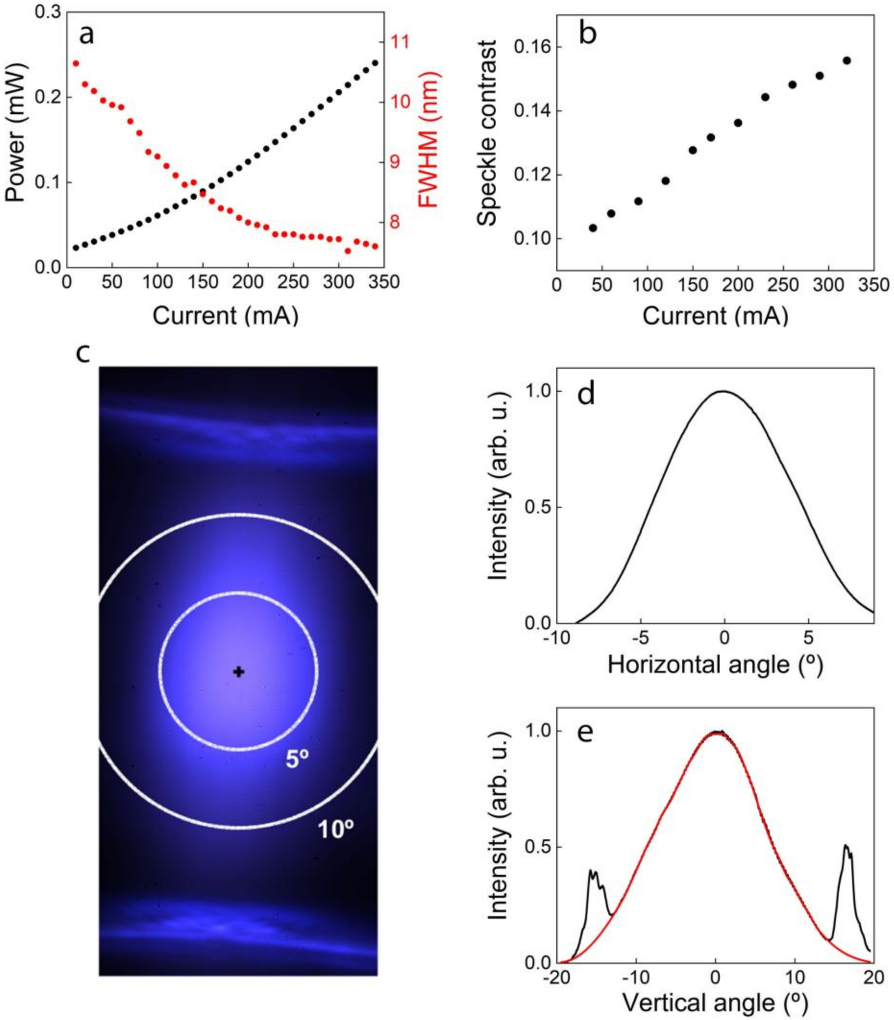




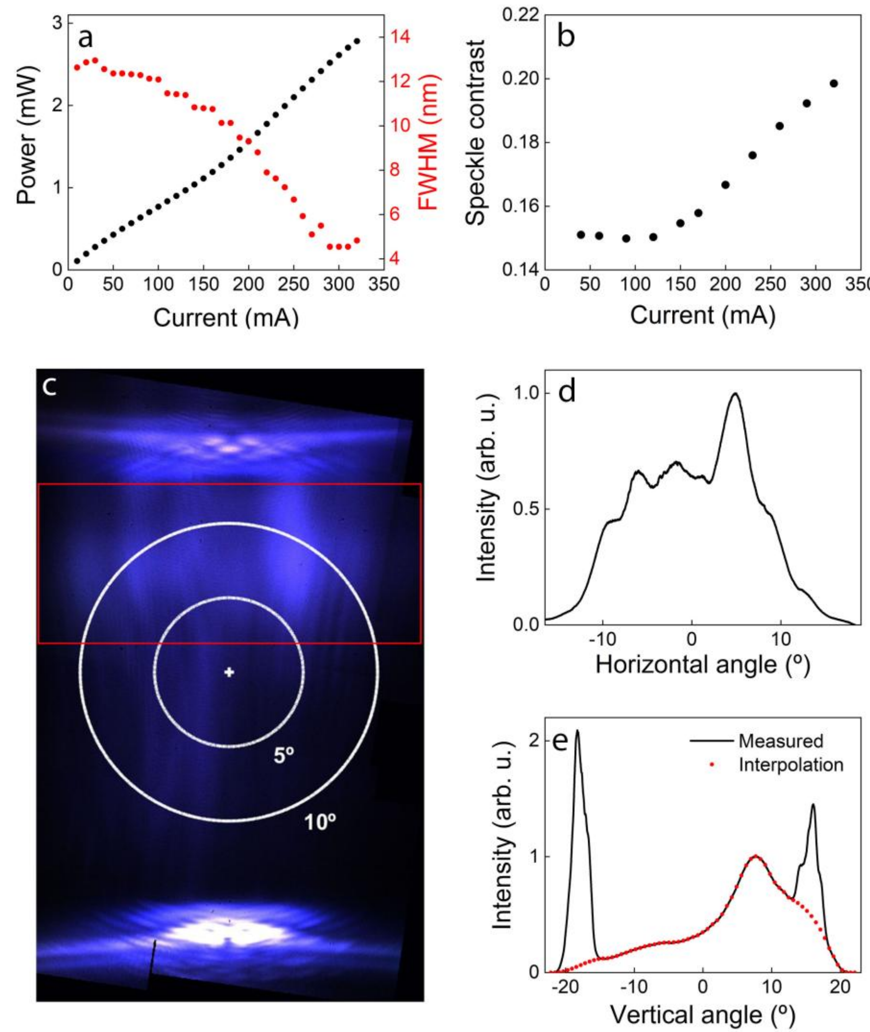
A. M. Pacheco, A. Consoli and C. López "Feedback suppression in 405 nm superluminescent diodes via engineered scattering" J. Phys. Photonics 8 015059 (2026)

Results: devices B and C

Device B



Device C



Device comparison

Device	Max output (mW)	Min linewidth (nm)	Max contrast	div angles (hor)	div angles (ver)
A	1.8	5.9	0.2	12	35
B	0.24	7.5	0.16	14	35
C	2.8	5	0.2	25	38

Table S1. Comparison of emission characteristics of devices A, B and C.

- Longer amplification length, higher output
- Scattering defects on the output facet, detrimental for beam quality and divergence
- Lower spatial coherence than original LD in all devices

Conclusions

Technique demonstration:
Demonstration of feedback suppression in blue LDs by introduction of scattering defects via pulsed laser ablation

Comparison of devices:
The best device shows broad spectrum (~6nm) and limited output power (~2mW), however enough for OCT and FOG applications.

Cost reduction:
The added value of the method is the reduction in fabrication cost, compared to ex-novo SLD realization, due to post-processing a cheaper (5x to 10x) device.

Thank you

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